

Internship in Physical Organic Chemistry

Week 1 — Part 4: Writing the Background chapter

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Goals:

- Build an argument outline from your reading templates before writing any prose
- Write a complete draft of the Background chapter, one paragraph per claim
- Revise the draft by checking structure first, then sentences
- Produce a chapter that synthesises the literature rather than summarising it paper by paper

In Part 3 you read your selected papers carefully and recorded what they show, what they claim, and how they relate to each other. In this part you will turn those notes into a written Background chapter.

The single most important principle in this part is: **write the argument before you write the text**. This means producing a skeleton of claims — one sentence per paragraph — before you write a single sentence of prose. Every experienced scientific writer works this way, even if they do not call it that. Writers who open a blank document and start writing prose from the top of a page typically produce summaries, not arguments. The goal of a Background chapter is not to report what each paper says. It is to construct the conceptual foundation that makes your own experiment necessary and intelligible.

1 Phase A: building the outline

1.1 What a Background chapter must do

A Background chapter in a scientific report has one job: to bring the reader from what they already know to the precise point where your experiment begins. It does this by building an argument, not by reviewing the literature.

The distinction matters. A *summary* of the literature reports what papers exist and what they found. An *argument* uses the literature to establish a set of claims that lead logically to a question that only your experiment can answer.

A well-structured Background chapter typically does three or four things in order:

1. Introduces the chemical system or reaction and explains why it is interesting from the perspective of physical organic chemistry.
2. Describes what is already known about its behaviour — the observations and interpretations that are established in the literature.
3. Identifies what is not yet known, or what remains uncertain or contested.
4. States (or implies) the question that follows from this gap.

This is the architecture of your outline. You do not need to follow it rigidly, but every Background chapter must accomplish these four things in some order.

1.2 Writing your claim sentences

Open your completed reading templates and your answers to the three comparison questions from Part 3 (what papers agree on, where they differ, what is missing). These are your raw material.

Task. Working from your templates and comparison notes, write three to four claim sentences. Each sentence will become the topic sentence of one paragraph in your Background chapter.

A claim sentence has a specific form. It is not a description of a paper (“Smith et al. showed that...”). It is a statement about the chemistry (“The Diels-Alder reaction proceeds faster in polar solvents than in non-polar ones.”). The literature supports the claim; the claim is not a report of the literature.

Write your claim sentences without looking at the papers. Use only your templates. If you cannot write a claim without consulting the paper again, your template notes are not yet detailed enough — go back and add to them.

Illustrative example. Imagine a student working on a project about the solvent dependence of a substitution reaction. After reading their selected papers and completing their templates, they produce the following four claim sentences:

1. S_N2 reactions proceed via a single concerted step in which bond formation and bond breaking occur simultaneously, with the transition state carrying a partial negative charge distributed between the nucleophile and the leaving group.
2. The rate of S_N2 reactions increases with solvent polarity because polar solvents stabilise the charged transition state relative to the neutral ground state.
3. Protic solvents, however, retard S_N2 reactions by solvating the nucleophile and reducing its effective reactivity, an effect that is absent in aprotic polar solvents.
4. The relative contribution of transition-state stabilisation and nucleophile solvation to the observed rate has not been separated experimentally for the specific substrate class studied here.

These four sentences, read in order, establish what the reaction is, why solvent matters, why the effect is more complex than it first appears, and what specific question remains open. That progression is the architecture of the Background chapter — and none of it required mentioning a single author’s name.

1.3 Checking the logic of your outline

Before writing any prose, read your claim sentences aloud, in order. Ask yourself three questions:

1. Does each sentence follow logically from the one before it? A reader who knows nothing about your topic should be able to follow the progression.
2. Does the last sentence point clearly toward your experiment? If someone read only your claim sentences, would they understand why your project exists?
3. Is any claim unsupported by your reading? If you cannot point to specific evidence in your templates for a claim, either remove the claim or go back to the literature.

Task. Read your claim sentences in order and apply the three checks above. Adjust the order or wording until the sequence is logically coherent. You may have three sentences or five — the number matters less than the logic. Do not proceed to Phase B until your outline passes all three checks.

This step typically takes 15 to 30 minutes. It is not wasted time. Every adjustment you make to the outline now prevents a structural rewrite of the prose later.

2 Phase B: drafting the paragraphs

2.1 The structure of a scientific paragraph

Each paragraph in your Background chapter corresponds to one claim sentence from your outline. The claim sentence becomes the **topic sentence** — the first sentence of the paragraph. Everything else in the paragraph supports, illustrates, or qualifies that claim.

A well-formed paragraph in scientific writing has the following structure:

1. **Topic sentence.** States the claim. Appears first, without preamble. The reader should know immediately what this paragraph is about.
2. **Supporting sentences.** Two to four sentences that provide evidence, examples, or elaboration. Each supporting sentence must be directly relevant to the topic sentence. If a sentence could belong to a different paragraph, it does not belong here.
3. **Closing sentence.** One sentence that either summarises the paragraph's point or creates a logical connection to the next paragraph. This sentence should not introduce new information.

On length. A paragraph in scientific writing is typically four to seven sentences long. Shorter paragraphs are appropriate when the claim is simple and well-established. Longer paragraphs risk losing coherence. If you find yourself writing eight or more sentences, ask whether the paragraph is making one claim or two. If two, split it.

2.2 Writing from your templates, not from the papers

When you write the supporting sentences of each paragraph, work from your reading templates, not by going back to the papers and paraphrasing them. Your templates contain the observations and interpretations you extracted in Part 3. These are your evidence.

This is not a shortcut — it is the correct method. Going back to papers while drafting encourages you to copy their structure and language. Your paragraph should be built around your claim, not around the organisation of a paper you are citing.

Task. For each claim sentence in your outline, write one complete paragraph. Proceed in this order:

1. Write the topic sentence exactly as it appears in your outline, or improve it slightly if needed.
2. Look at your reading templates and identify the two or three pieces of evidence that best support this claim. Write one sentence per piece of evidence.
3. Write a closing sentence.
4. Move to the next claim.

Write all paragraphs before revising any of them. Do not edit while drafting.

2.3 Setting up citations: from Zotero to Overleaf

Before you can cite anything in LaTeX, you need to export your references from Zotero and make them available to your Overleaf project. This is a one-time setup that takes about five minutes.

Step 1: export your Zotero collection as a BibTeX file

In Zotero, right-click on your `week1_literature` collection in the left panel and select **Export Collection**. In the dialogue that appears, choose **BibTeX** as the format. Make sure the option *Export Files* is **not** ticked — you only need the metadata, not the PDFs. Click OK and save the file. Name it `references.bib`.

Important: citation keys. Each entry in the `.bib` file has a unique key that you will use in your `\cite{}` commands. Zotero generates these keys automatically, typically in the form `AuthorYEARword`, for example `Smith2001fragmentation`. Before uploading, open the `.bib` file in a text editor and check that each key is readable and unique. If two entries have the same key, one will silently overwrite the other.

Step 2: upload the file to Overleaf

In your Overleaf project, click the **Upload** button (the arrow icon at the top of the file list on the left). Select your `references.bib` file. It should now appear in the file list.

Step 3: connect the bibliography to your document

Open your main `.tex` file. At the point where you want the reference list to appear (typically at the end of the document, before `\end{document}`), add these two lines:

```
\bibliographystyle{plain}
\bibliography{references}
```

The first line sets the citation style. The second line tells LaTeX which `.bib` file to use (without the `.bib` extension). Click **Recompile**. If everything is set up correctly, a reference list will appear at the end of the PDF.

Step 4: inserting citations in the text

Once the bibliography is connected, you can cite any entry using its key. In LaTeX, a citation is inserted with `\cite{key}` and appears at the end of the sentence, before the full stop:

The reaction rate increases with solvent polarity \cite{Smith2001}.

To cite more than one paper in the same location, separate the keys with a comma and no space:

This has been observed in several compound classes \cite{Smith2001,Jones2005}.

Keeping references up to date

Each time you add a new paper to Zotero, you must re-export the collection and re-upload the `.bib` file to Overleaf, replacing the old one. The simplest habit is to do this at the end of each working session, not in the middle of writing. If you cite a key that is not yet in the `.bib` file, LaTeX will compile without error but will show `[?]` in the PDF instead of the citation.

Two conventions that apply throughout scientific writing:

- Cite the *source of the claim*, not the most famous paper on the topic. If you learned something from a specific paper, cite that paper.
- Do not cite a paper you have not read. If you found a claim in a review that cites an original paper, read the original paper before deciding whether to cite it.

On the active voice. Scientific writing in English uses the active voice whenever possible. Compare:

Passive: “Fragment ions at m/z 153 and 129 were observed in the spectrum.”

Active: “The spectrum shows fragment ions at m/z 153 and 129.”

The active version is shorter and clearer. The subject performs the action. Use the passive voice only when the agent is genuinely unknown or irrelevant.

3 Phase C: revising the draft

Once you have a complete draft, you revise it in two passes. The first pass checks structure. The second pass checks sentences. Do not mix them.

3.1 First pass: structure

Read only the first sentence of each paragraph, in order. These are your topic sentences. They should form a coherent argument on their own, without the supporting sentences.

Task. Copy only the first sentence of each paragraph into a new Overleaf document or a blank page. Read them in sequence and ask:

- Does the progression make sense? Could a reader follow the argument from these sentences alone?
- Does any topic sentence seem out of place? If so, move the whole paragraph.
- Does any topic sentence repeat the claim of another? If so, merge the two paragraphs or delete one.
- Does the last topic sentence connect to what your experiment measures? If not, add a transition sentence or adjust the final claim.

Make structural changes before reading the supporting sentences. Revising sentences inside a paragraph that is in the wrong place is wasted effort.

3.2 Second pass: sentences

Once the structure is correct, read the full draft and apply the following checks to each sentence:

- **Cut unnecessary words.** Every word that does not add meaning should be removed. “It is important to note that...” adds no information. “It has been shown by Smith et al. that...” can almost always be replaced by “Smith et al. showed that...” or, better, by a direct statement of the claim followed by the citation.
- **Place the main information early.** The subject of a sentence should be the thing the sentence is about. If you find yourself writing long introductory clauses before the main verb, rewrite.
- **Use strong verbs.** Prefer “shows”, “produces”, “yields”, “establishes” over “is seen to show”, “can be considered to produce”.
- **Check that each sentence is either observation or interpretation**, not both. If a sentence contains a measurement and a mechanistic claim in the same clause, split it into two sentences.

What revision is not. Revision is not proofreading. Proofreading fixes spelling and punctuation, and happens last. Revision changes what you are saying and how you are organising it. If you proofread before revising, you will proofread text that you later delete.

4 What your Background chapter is not

The most common error at this stage is writing a Background chapter that reads as a series of paper summaries:

Smith et al. (2001) studied the fragmentation of flavonoids using MS/MS. They found fragment ions at m/z 153 and 127. Jones et al. (2005) also studied flavonoid fragmentation. They reported...

This is not a Background chapter. It is an annotated bibliography. The reader learns what papers exist but not what the field has established.

A Background chapter makes *claims about the chemistry* and uses papers as evidence for those claims. The same information, written correctly, looks like this:

Under low-energy CID conditions, the C ring of flavonoids undergoes retro-[4+2] cycloreversion, producing fragment ions at m/z 153 and 127 that are diagnostic for the A- and B-ring substitution pattern [?, ?]. This fragmentation pathway is well established across the flavonoid subclasses and forms the basis for structural assignment in untargeted metabolomics workflows [?].

The chemistry is in the foreground. The papers are in the background, as citations. This is the correct relationship between argument and evidence.

5 End of Week 1 checklist

Before submitting your Background draft, verify the following:

- Your Background chapter has a clear outline: three to four paragraphs, each built around a single claim.
- Each paragraph begins with a topic sentence that states a claim about the chemistry, not about a paper.
- Every factual claim is supported by a citation.
- You have completed the structural revision pass (topic sentences read coherently in sequence).
- You have completed the sentence revision pass (active voice, no clutter, observations and interpretations separated).
- Your Zotero library is up to date and linked to Overleaf.
- You can say in two sentences what your Background chapter establishes and why it makes your Week 2 experiment necessary.